

infrastructure, cost, technology used, quality and so on.

The solar PV value chain extends from production of polysilicon to the solar PV system. Polysilicon is the first raw material in the value chain, which is used to produce wafers. Wafers are processed into cells, which are assembled together to create modules, either of the crystalline silicon (c-Si) or thin-film variety. The photovoltaic system typically includes an array of solar panels, an inverter, and sometimes a battery and interconnection wiring.

As of now, it's mainly the modules and, to some extent, cells which are being manufactured in India. Solar wafer manufacturing is yet to witness growth.

"The wafer manufacturing capacity in India is less than 20 MW currently, while the requirement at the end of this decade will be 2000 MW per year. It has not taken off primarily due to the high capital investments. Wafer and cell manufacturing requires high capital-intensive investments—a minimum of \$50-75 million for wafer manufacturing and about \$20 million for cell manufacturing vis-a-vis about \$1-2 million required to put up a module making facility. Consequently, there has been a strong momentum behind setting up module plants," informs Rajesh Menon, deputy direc-

Industry at a glance

1. India is amongst top five destinations worldwide for solar energy.
2. Currently, a nascent and evolving industry, this segment has gained considerable momentum over the last two years with the launch of the Jawaharlar Nehru National Solar Mission (JNNSM), which entails provision of generation-based incentives to solar power developers and other incentives like renewable energy incentives.
3. Under the JNNSM, India is expected to reach a total solar installed capacity (both PV and CSP) of 20,000 MW and a total electricity generation of 25-30 TWh per year by 2022.
4. The top five states in India for solar PV potential are Gujarat, Rajasthan, Maharashtra, Tamil Nadu and Andhra Pradesh, respectively.
5. Gujarat has witnessed the fastest ramp up in solar capacity over the last two years. It accounts for about 50 per cent of the installed solar capacity in India.
6. As of now, it's mainly the modules and, to some extent, cells which are being manufactured in India.
7. Currently, the wafer manufacturing capacity in India is less than 20 MW, while the requirement at the end of this decade will be 2000 MW per year.
8. Going forward, we may see polysilicon and wafer manufacturing plants coming up in India, with companies like Lanco Solar, Bhaskar Silicon, Carborundum Universal and Yash Birla Group announcing their plans.
9. Several companies have made a foray into the sector, including Azure Power, Camelot Enterprises and SunEdison. On the equipment front as well, the solar potential in the country has attracted numerous foreign solar global majors with First Solar investing more than \$500 million to set up a thin-film CdTe manufacturing plant near Chennai.
10. Solar and thermal power will cost same in five years. India has advanced the target date for selling solar power at the same rate as conventional electricity to 2017 as tariffs have fallen significantly in the latest projects on offer and the trend is likely to continue.

tor general, Confederation of Indian Industry (CII).

However, the majority of the modules produced in India so far have catered to the export market, in the absence of domestic demand. Until as recently as 2010, there was no domestic solar market in the country, which meant that cell and module manufacturers had to rely primarily on exports for their production.

Unfortunately, today, when demand is there with more and more solar PV power plants coming up across the country, the market for domestic manufacturers is marred by stiff competition from key global manufacturers including Shin-Etsu, Sanyo and Sumko from Japan, MEMC from USA, LDK and Renesola from China, and Topsil from Denmark. Many of these companies have years of

experience and considerable scale as of today.

Also, there are many factors affecting the cost of solar cells made in India, putting these at a disadvantage. First, the raw material and components for solar cells still have to be imported to some degree—like the vital silicon wafers—which raises the price of solar cells in India.

Currently, India is not strong in the manufacture of silicon as it requires good infrastructure like quality power at low cost (around Rs 2.5/kWh (per unit)). Going forward, though, we may see polysilicon and wafer manufacturing plants coming up in India, with companies like Lanco Solar, Bhaskar Silicon, Carborundum Universal and Yash Birla Group announcing their plans. Lanco plans to set up the facility in Raipur with power coming from its thermal power plant in Chhattisgarh.

Most cell manufacturers in India are using the first-generation crystalline silicon (cSi) technology, which is as of now the most efficient but also the most expensive. This keeps the costs of cells made in India at a higher level

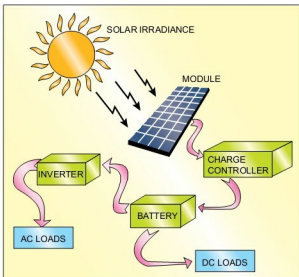


Fig. 1: Working of solar PV system